endosomal maturation (Gilleron et al., 2013). Quantitative approaches such as nanoscale secondary ion mass spectrometry (NanoSIMS) also provided high-resolution spatial and absolute measurements of oligonucleotide distribution and confirming dense accumulation within endosomal structures and minimal cytosolic presence; again, yielding estimates of ~1–2% productive escape in vivo (He et al., 2021; Dowdy, 2023). Biochemical fractionation methods, which physically separate cytosolic and membrane-bound compartments, further support these findings; despite a known bias toward overestimating cytosolic content due to endosomal disruption during lysis, these studies still report similarly low cytosolic fractions (~1%) (Juliano, 2018). Live-cell fluorescence imaging offers mechanistic insight into this inefficiency, demonstrating that escape is rare and transient (Wittrup et al., 2015).

Functional assays provide an additional layer of validation, as only cytosolic siRNA can engage the RNA-induced silencing complex; the observation that relatively few intracellular molecules are sufficient for robust gene knockdown implies that only a small proportion of the total internalized pool is productively delivered (Wittrup et al., 2015; Dowdy, 2023). Finally, in vivo pharmacokinetic and pharmacodynamic analyses, particularly of GalNAc–siRNA conjugates, reinforce this paradigm, indicating that only 0.3% of internalized siRNA is present in the cytosol at steady state, reflecting slow and continuous leakage from endosomal compartments rather than efficient release (Brown et al., 2020; Dowdy, 2023). Collectively these efforts provide strong evidence that virtually all internalized RNA remains sequestered within endosomal compartments and escape as the rate-limiting step in siRNA delivery.

To address this delivery bottleneck, a wide range of strategies have been developed that aim to enhance endosomal escape by transiently destabilizing membranes or altering intracellular trafficking pathways. Small-molecule agents like chloroquine and other cationic amphiphilic compounds can act by buffering endosomal acidification and inducing osmotic swelling, membrane perturbation, or vesicle rupture, thereby facilitating release of entrapped cargo (Grau and Wagner, 2024; Du Rietz et al., 2020). However, these approaches are inherently non-specific, often disrupting multiple endo-lysosomal compartments, and are limited by dose-dependent cytotoxicity and poor translational performance in vivo, where concentrations required for efficient escape are typically not well tolerated.

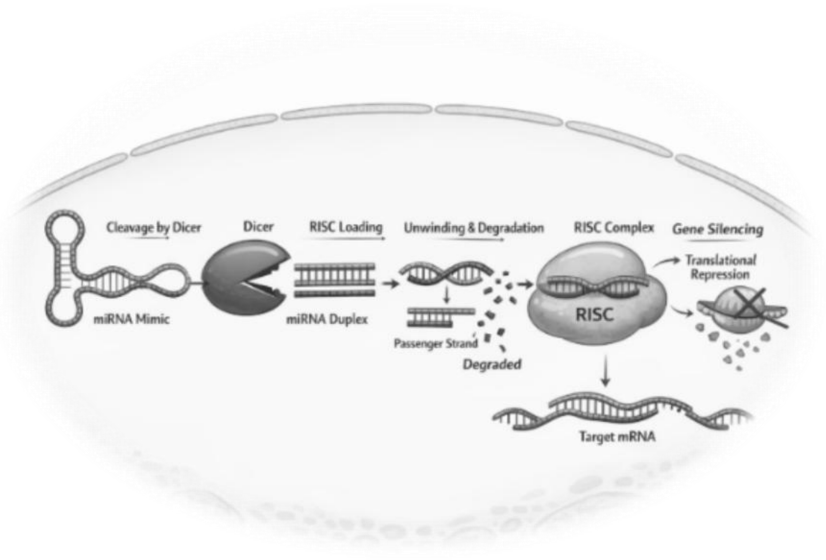
Peptide-based approaches such as arginine-rich cell-penetrating peptides and pH-responsive peptides exploit electrostatic interactions and acid-triggered conformational changes to promote membrane fusion, pore formation, or leakage from late endosomal compartments (Grau and Wagner, 2024; Mastrobattista et al., 2002; Wadia et al., 2004), but these systems frequently suffer from serum instability and immunogenicity, as membrane-lytic activity can extend beyond endosomes to other cellular membranes.

Nanoparticle-based delivery systems, particularly ionizable lipid nanoparticles and cationic nanoparticles, are designed to exploit the endosomal environment by undergoing protonation at low pH and interact with anionic lipids to promote formation of non-bilayer structures that destabilize membranes and enable cytosolic release (Cullis and Hope, 2017; Hafez et al., 2001). However, despite their clinical success, these modalities retain intrinsically low escape efficiencies, are largely biased toward hepatic tissue, and can elicit inflammatory or infusion-related responses. These limitations are reflected in the strategic shift by several companies toward ligand–conjugate approaches like GalNAc. For example, Alnylam's transition away from lipid nanoparticle–formulated siRNA (patisiran/Onpattro) toward GalNAc-conjugated platforms was driven by the need for more predictable pharmacokinetics, improved tolerability, and simpler subcutaneous dosing; therapeutic compatibility and commercial viability often relegate endosomal escape strategies to a lower priority.

Additional approaches, including photochemical internalization and emerging chemical biology strategies that modulate endosomal trafficking machinery, further underscore the breadth of mechanisms being explored to overcome this barrier (Grau and Wagner, 2024; Juliano, 2021). However, these strategies are often constrained by practical limitations, including restricted tissue penetration and the inherent complexity and consequences of perturbing intracellular trafficking networks. These challenges highlight that enhancing endosomal escape is not simply a question of increasing membrane disruption, but rather a fundamentally constrained optimization problem that requires balancing delivery efficiency with cellular integrity and therapeutic compatibility. Despite extensive innovation across multiple modalities, substantial improvements beyond the low baseline escape efficiency observed across systems have remained elusive.

	ASO (Antisense Oligonucleotides)	siRNA (non-conjugated / baseline)
Absorption (SC → plasma)	Rapid absorption via lymphatics; Tmax 2–6 h *(Geary et al., 2015)*	Rapid absorption; Tmax 1–4 h, but often lower bioavailability *(Ayyar & Song, 2023)*
Plasma exposure	Moderate Cmax; declines within hours (≤24 h) due to rapid tissue uptake *(Geary et al., 2015)*	Low and transient plasma exposure; often cleared within hours *(Ayyar & Song, 2023)*
Primary tissues (SC dosing)	Liver, kidney, spleen; broad distribution via protein binding *(Geary et al., 2015)*	Limited distribution without targeting; some liver/kidney uptake but inefficient *(Ayyar & Song, 2023)*
Time to tissue distribution	Rapid uptake into tissues within hours post-dose *(Geary et al., 2015)*	Within hours, but lower fraction reaches productive intracellular compartments
Tissue half-life	Long: 1–4 weeks (liver/kidney) due to intracellular retention *(Geary et al., 2015)*	Functional half-life depends on RISC; days–weeks if successfully loaded, otherwise short
Intracellular trafficking	Endosomal uptake → gradual release; relatively higher intracellular accumulation	Endocytosis → endosomal escape is limiting (<1% escape) *(Ayyar & Song, 2023)*
Onset of PD effect	Typically, 1–3 days post-dose (depends on RNA turnover)	Typically 2–7 days post-dose (processing + RISC loading + RNA turnover)
Peak PD effect	Often ~3–7 days post-dose	Often ~5–10 days post-dose
Duration of PD effect	Sustained for 1–4+ weeks depending on tissue retention *(Geary et al., 2015)*	Sustained for weeks–months if sufficient RISC loading occurs *(Hammond, 2005)*
PK–PD relationship	More tissue concentration-driven (liver/kidney levels correlate with effect) *(Geary et al., 2015)*	Decoupled: small intracellular active pool drives prolonged effect *(Hammond, 2005)*
Efficiency of tissue use	Higher fraction of dose becomes pharmacologically active	Very small fraction (<1%) becomes active (RISC-loaded)

Design Strategies & Delivery



miRNA mimics

A microRNA (miRNA) mimic is typically designed as a synthetic double-stranded RNA that resembles the endogenous miRNA duplex. It contains a guide strand (Antisense) corresponding to the mature miRNA sequence and a complementary passenger strand (Sense), forming a short duplex with characteristic features such as 2-nucleotide 3′ overhangs generated by Dicer processing. After introduction into the cell, the miRNA mimic duplex is unwound during RISC loading, the passenger strand is degraded, and the guide-strand is incorporated into the RNA-induced silencing complex (RISC). Within RISC, the guide strand directs gene silencing by partial base pairing with target mRNAs, leading to translational repression or a catalytic mRNA degradation.

miRNAs are powerful regulators that can control multiple genes and entire cellular pathways, making them promising targets for therapy as well as biomarkers. However, despite strong preclinical progress and growing interest, especially following the success of mRNA vaccines, the field remains in early clinical stages, with no FDA-approved miRNA therapies and several trials halted due to toxicity and immune-related side effects. It remains unclear whether the observed toxicities in miRNA therapeutics arise from the miRNA or the delivery system.

MRX34 was a first-in-class microRNA therapeutic composed of a synthetic double-stranded miR-34a mimic formulated within a lipid nanoparticle delivery system to restore the tumor-suppressive miR-34 pathway downstream of p53 and repress multiple oncogenic targets. Preclinical studies and early Phase 1 clinical data demonstrated successful systemic delivery, target engagement, and limited antitumor activity, supporting proof-of-concept for miRNA replacement therapy. However, the trial was terminated in 2016 due to severe immune-related toxicities, including cytokine release syndrome and treatment-related deaths, likely caused by innate immune activation triggered by the nanoparticle. This outcome highlighted key challenges in RNA therapeutics particularly the balance between structural design, delivery efficiency, and immunogenicity despite the strong biological rationale for miR-34–based therapy (Zhang et al., 2019; Hong et al., 2020)

MRX34 is an unmodified RNA formulated using a liposomal nanoparticle delivery system (NOV340) composed of amphoteric lipids designed to encapsulate and protect the double-stranded miR-34a mimic during systemic administration. The lipid formulation typically included components such as phospholipids, cholesterol, and PEGylated lipids to enhance stability, prolong circulation time, and facilitate cellular uptake and endosomal release of the RNA cargo.

In the Phase 1 clinical trial, MRX34 was administered intravenously twice weekly (commonly on days 1 and 4 of a 3-week cycle), with dose escalation up to a maximum tolerated dose of approximately 110 mg/m² (about 40mg per dose) in non-hepatocellular carcinoma and HCC patients, before immune-related toxicities limited further dosing (Hong et al., 2020; Beg et al., 2017). The trial was terminated due to serious immune-mediated toxicities, including fatal adverse events in multiple patients (Kim and Croce, 2023) . These toxicities were characterized by strong immune activation and while MRX34 successfully downregulated oncogenic target genes, the severity of these adverse effects led to trial discontinuation (Kim and Croce, 2023).

The toxicity and clinical outcomes observed with MRX34 cannot be attributed to a single factor, as multiple confounding elements were involved. The lipid nanoparticle delivery system facilitated broad systemic distribution, potentially exposing diverse tissues to the therapy, while frequent and high dosing likely led to excessive off-target accumulation. The complexity of the drug product complicates efforts to determine whether toxicity was primarily driven by the delivery system, payload miRNA, or the intrinsic biology of miRNA regulation.

An alternative to traditional, repeated systemic delivery of synthetic miRNA mimics and lipid nanoparticles (LNPs) involves a vector-mediated strategy for sustained, intracellular expression. Researchers at UCL (University College London) have utilized viral vector gene therapy to achieve this, specifically through the AAV5-delivered miRNA construct underlying AMT-130. This approach employs an engineered pri-miR-451a scaffold to express an anti-huntingtin microRNA directly within transduced neurons.

Unlike MRX34, which required frequent intravenous dosing due to the transient nature of synthetic RNA, the AAV-mediated delivery enables long-term, endogenous production of the therapeutic miRNA following a single administration. In this approach, the therapeutic sequence is not delivered as a mature, fragile duplex RNA; it is instead delivered as a DNA cassette encoding a pri-miRNA transcript, which is subsequently processed by the host native machinery into the active guide strand.

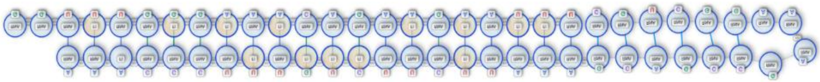
AMT-130 incorporates a CAG promoter, engineered pri-miR-451 flanking regions, a pre-miRNA hairpin containing the HTT-targeting guide strand, and an hGH polyadenylation signal cassette to support stable neuronal expression and accurate RNA processing (Evers et al., 2018). Importantly, the miR-451 scaffold was selected because it undergoes Ago2-dependent, Dicer-independent maturation, reducing passenger-strand accumulation and potentially improving strand specificity compared with conventional shRNA or duplex miRNA approaches (Cheloufi et al., 2010; Yang et al., 2010). Although AAV-mediated delivery introduces its own challenges particularly the permanence of transgene expression, limited reversibility, and concerns regarding long-term safety and immune responses to viral capsids, it avoids several limitations associated with systemic nanoparticle delivery, including repeated high-dose exposure, broad biodistribution of synthetic RNA, and dose-limiting innate immune activation. Consequently, the transition from MRX34-like nanoparticle systems to AAV-expressed miRNA therapeutics reflects a broader evolution in the RNA therapeutics field toward more durable, tissue-targeted, and biologically integrated delivery platforms.

The therapeutic cassette described in the patent consists of a fully engineered adeno-associated viral (AAV) expression designed to achieve long-term suppression of huntingtin (HTT) expression in neurons through RNA interference. In the disclosed pVD-CAG-miH12-451 configuration, the cassette begins with the synthetic CAG promoter, a strong hybrid promoter combining the cytomegalovirus (CMV) enhancer and chicken β-actin promoter elements, included to drive high-level transgene expression in neuronal tissue following intracerebral AAV delivery (Evers et al., 2018). Downstream of the promoter is the 5′ pri-miRNA flanking region derived from endogenous miR-451a. These flanking sequences are necessary because they provide the structural and sequence context recognized by the endogenous microRNA-processing machinery, particularly the Drosha/DGCR8 complex, enabling correct cleavage of the primary transcript into a precursor miRNA hairpin (Cheloufi et al., 2010). Embedded within this framework is the engineered pre-miRNA hairpin itself, which constitutes the active therapeutic silencing element. The hairpin contains the HTT-targeting guide strand sequence 5′-AAGGACUUGAGGGACUCGAAG-3′ (SEQ ID NO.5), which is fully complementary to the selected exon-1 huntingtin target sequence and mediates sequence-specific degradation of HTT mRNA through incorporation into the RNA-induced silencing complex (RISC) (Evers et al., 2018). Opposite the guide strand is the engineered passenger strand, designed not only to stabilize the stem-loop structure but also to minimize passenger-strand loading into RISC and thereby reduce off-target silencing effects. The use of the miR-451a scaffold is particularly important because miR-451 follows a non-canonical Ago2-dependent, Dicer-independent maturation pathway that preferentially produces a single dominant guide strand, improving strand specificity and reducing unintended RNAi activity (Cheloufi et al., 2010; Yang et al., 2010). Following the pre-miRNA region, the cassette contains the 3′ pri-miRNA flanking sequence, which further supports accurate Drosha processing, transcript stability, and nuclear export of the precursor hairpin. Finally, the construct terminates with the human growth hormone (hGH) polyadenylation signal, which directs efficient transcriptional termination and polyadenylation, thereby stabilizing the therapeutic transcript after expression from the AAV genome. Collectively, these engineered sequence components form a highly optimized neuronal RNAi cassette intended to provide durable expression, efficient miRNA biogenesis, selective guide-strand loading, and sustained suppression of mutant and wild-type huntingtin expression within the central nervous system (Evers et al., 2018).

The role of Dicer

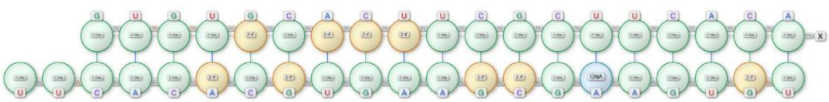
RNA interference (RNAi) is a highly conserved mechanism for post-transcriptional gene regulation. In mammalian systems, this process is mediated by a series of coordinated molecular interactions involving the RNase III enzyme Dicer, which cleaves longer dsRNA into short interfering RNAs (siRNAs), and the RNA-induced silencing complex (RISC); uses the guide strand to recognize and cleave target transcripts (Hefner et al., 2008).

While standard (Alnylam-style) synthetic 21–23 nucleotide siRNAs mimic the “products” of Dicer processing and can directly engage RISC, they bypass the natural Dicer processing step. This may lead to reduced efficiency and less controlled strand selection. In contrast, longer Dicer-substrate RNAs (e.g., 27mers), designed to engage the endogenous RNAi machinery more fully, may enable Dicer-mediated cleavage and directional handoff to RISC and potentially leading to enhances potency, durability, and specificity of gene silencing (Hefner et al., 2008).   This is the theory behind the platform technology developed by a company called Dicerna which sold to Novo Nordisk acquired in 2021 for $3.3 billion.



A research group from Bio-Rad Laboratories in collaboration with Integrated DNA Technologies (IDT) performed a series of experiments comparing Dicer-substrate 27mer RNAs with conventional 21mer siRNAs to evaluate the impact of engaging the Dicer processing step

on RNAi performance. In HeLa cell models across multiple gene targets (including CDK2, TP53, RAF1, and ACTB), the Dicer-substrate 27mers produced greater gene knockdown in dose–response studies, particularly at lower concentrations, indicating enhanced intrinsic potency (Hefner, E., Clark, K., Whitman, C., Behlke, M.A., Rose, S.D., Peek, A.S. and Rubio, T. (2008). They also demonstrated more sustained silencing over time, whereas the activity of 21mer siRNAs declined more rapidly. The authors attributed these improvements to Dicer-mediated processing and efficient handoff to RISC, which promotes accurate guide-strand selection and improved complex loading. Notably, the ability to achieve effective silencing at lower doses suggests a potential reduction in off-target effects and toxicity, supporting both improved efficacy and safety relative to traditional siRNA designs (Hefner et al., 2008).



siRNA

The typical design of small interfering RNAs (siRNAs) as 21–23 nucleotide double-stranded duplexes with 2-nucleotide 3′ overhangs is considered optimal because it closely mimics the natural products of Dicer processing and maximizes compatibility with the RNA interference (RNAi) machinery. Elbashir et al. (2001) demonstrated that synthetic 21-nt siRNA duplexes are sufficient to induce efficient and sequence-specific gene silencing in mammalian cells while avoiding the nonspecific responses triggered by longer double-stranded RNAs. The presence of 2-nt 3′ overhangs, typically composed of UU or TT residues, further enhances silencing efficiency by facilitating Dicer recognition and promoting effective loading into the RNA-induced silencing complex (RISC), as these structural features resemble endogenous Dicer cleavage products. Additionally, the requirement for a double-stranded RNA duplex is critical for RISC assembly, as strand selection and guide strand incorporation depend on duplex structure and thermodynamic asymmetry (Sioud, 2025).

21–23 nucleotides
2-nt 3′ overhangs (usually “UU” or “TT”)
Double-stranded RNA duplex required for RISC loading

Strand Selection and Thermodynamic Asymmetry

Efficient siRNA function depends on strand selection driven by thermodynamic asymmetry within the duplex. Specifically, the 5′ end of the antisense (guide) strand is designed to have lower thermodynamic stability, while the 5′ end of the sense (passenger) strand is relatively more stable, creating an asymmetry that directs RISC to preferentially incorporate the correct guide strand (Schwarz et al., 2003; Sioud, 2025). This asymmetry is further reinforced by nucleotide composition, where an A/U-rich region at the 5′ end of the antisense strand reduces stability, and a G/C-rich region at the 5′ end of the sense strand increases stability (Ui-Tei et al., 2004). Together, these features promote accurate strand selection, ensuring efficient guide strand loading into RISC and enhancing overall gene silencing specificity and efficacy.

Lower stability at the 5′ end of antisense (guide) strand
Higher stability at the 5′ end of sense (passenger) strand
This promotes correct guide strand loading into RISC

A/U-rich at antisense 5′ end
G/C-rich at sense 5′ end

GC Content

Optimal siRNA efficiency is strongly influenced by GC content, with a moderate range of approximately 30–50% providing the best balance between stability and functionality. Excessively high GC content increases duplex stability, which can hinder unwinding by RISC and reduce silencing efficiency, whereas very low GC content weakens target mRNA binding and compromises activity (Reynolds et al., 2004). Consequently, functional siRNAs typically exhibit moderate GC levels combined with relatively lower internal stability, facilitating efficient strand separation and target recognition within the RNAi pathway (Sioud, 2025).

Optimal 30–50% GC
Too high GC leads to overly stable duplex and poor RISC unwinding
Too low GC results in weak target binding
Functional siRNAs tend to have moderate GC and lower internal stability

Nucleotide Preferences

In the antisense (guide) strand, the presence of A/U at position 1 (the 5′ end) reduces thermodynamic stability and promotes its incorporation into RISC, while a corresponding G/C-rich 5′ end in the sense (passenger) strand increases stability and supports proper strand discrimination (Ui-Tei et al., 2004). Additionally, the seed region (positions 2–8), which is critical for target mRNA recognition, tends to be more effective when A/U-rich, improving binding specificity and silencing efficiency (Reynolds et al., 2004). Conversely, long stretches of G/C bases or repetitive sequences are generally avoided, as they can increase internal stability and reduce siRNA functionality (Reynolds et al., 2004).

Antisense strand:
A/U at position 1 (5′ end)
Sense strand:
G/C at 5′ end

Seed Region

The seed region (positions 2–8) is crucial for target recognition and should ideally be A/U-rich while avoiding long GC stretches or repetitive sequences. Because this region largely determines binding specificity, it is also the main source of off-target effects: even partial complementarity can unintentionally silence other genes in a miRNA-like manner. To minimize this risk, sequences should be designed to avoid matching the 3′ UTRs of unrelated genes.

Seed (positions 2–8):
- Prefer A/U
- Avoid long GC stretches
- Main source of off-target effects
- Chemical modifications (e.g. 2′-O-methyl) can reduce off-target effects (Lennox and Behlke, 2016)

Target Site Considerations

Target sites should be accessible regions with minimal secondary structure, while also avoiding areas like the start codon and highly structured regions can hinder binding. Additionally, sequence design should exclude problematic motifs such as long repeats, palindromes, internal hairpins, and immune-stimulatory elements (e.g., GU-rich sequences) to ensure stability and reduce unwanted cellular responses.

Gapmer (RNAse-H) ASOs

Gapmer antisense oligonucleotides are single-stranded sequences typically 16–20 nucleotides long, designed with a central DNA “gap” (8–10 nt) flanked by chemically modified wings that enhance stability and binding (Bennett and Swayze, 2010; Crooke, 2017; Lennox and Behlke, 2016). The DNA gap is essential for forming an RNA–DNA hybrid, which recruits RNase H to cleave the target RNA. Chemical modifications such as LNA, 2′-O-methyl (2′-OMe), or 2′-O-methoxyethyl (2′-MOE) in the wings and a predominantly phosphorothioate (PS) backbone improve nuclease resistance, cellular uptake, and protein binding, while limited incorporation of phosphodiester linkages can fine-tune these properties without compromising activity (Kurreck, 2003; Bennett and Swayze, 2010; Geary et al., 2015). Effective gapmer design maintains 40–60% GC content and an appropriate melting temperature (55–65°C), targets accessible RNA regions, and avoids highly structured, repetitive, or protein-bound sites (Watts and Corey, 2012; Lennox and Behlke, 2016).

- GC content: 40–60%
- Melting temperature: 55–65°C
- Avoid:
  - Highly structured regions
  - Repetitive sequences
  - Protein-bound sites
  - Off-target & Toxicity Considerations
- Avoid sequence motifs:
  - CpG motifs
  - Long G runs
- Palindromes & Self-complementarity

To minimize off-target effects and toxicity, sequences should avoid motifs such as CpG, long G runs, palindromes, and self-complementarity, be screened for transcriptome-wide specificity, and be tested as multiple candidates (Krieg, 2006; Crooke, 2017; Bennett and Swayze, 2010). Gapmers function in both the nucleus and cytoplasm, leveraging RNase H's natural role in degrading RNA–DNA hybrids to achieve targeted gene silencing (Crooke, 2017; Wahba et al., 2011).

Steric-Blocking single-stranded ASOs

Steric-blocking antisense oligonucleotides are single-stranded sequences typically 15–25 nucleotides long with fully modified backbones, as they do not rely on RNase H activity (Bennett and Swayze, 2010; Lennox and Behlke, 2016). Instead, they function by physically blocking molecular interactions such as ribosome binding, spliceosome assembly, or protein–RNA interactions, enabling translation inhibition or splice modulation (Watts and Corey, 2012). These ASOs use fully modified chemistries such as 2′-OMe, 2′-MOE, LNA, or PMOs with phosphorothioate or neutral backbones to enhance stability, binding affinity, and cellular uptake (Kurreck, 2003; Summerton, 2007; Geary et al., 2015). Effective design requires moderate GC content (40–60%) and strong binding affinity, targeting accessible RNA regions such as start codons or splice regulatory elements while avoiding structured or protein-bound regions (Watts and Corey, 2012; Bennett and Swayze, 2010). To reduce toxicity and immune activation, sequences should avoid motifs like CpG, long G runs, and self-complementarity, and carefully balance high-affinity modifications (Krieg, 2006; Lennox and Behlke, 2016). These ASOs can function in both the nucleus and cytoplasm, with splice-switching applications primarily occurring in the nucleus.

Compare Splicing Technology: PMO vs 2'-MOE

	PMO (Morpholino; e.g., Eteplirsen, Golodirsen, Viltolarsen)	2′-MOE PS ASO (e.g., Nusinersen)
Backbone	Phosphorodiamidate (neutral)	Phosphorothioate (negatively charged)
Sugar/Scaffold	Morpholine ring (no ribose)	Ribose with 2′-MOE modification
Charge	Neutral	Negative
Protein binding	Low	High
Nuclease resistance	Very high	High
Binding affinity (Tm)	High	High–very high
Cellular uptake	Poor (often needs high dose or conjugates)	Better (PS-driven uptake)
Distribution	Limited without conjugates	Broad tissue distribution
Toxicity profile	Generally low	Moderate (PS-related effects possible)
Mechanism	Steric blocking (splice modulation)	Steric blocking (splice modulation)

Design implications: - PMOs prioritize safety and stability, with tradeoffs in delivery efficiency. MOE-PS ASOs balance delivery, potency, and pharmacokinetics, with increased biological interactions.

Conjugates and Delivery Strategies

RNA therapeutics have emerged as programmable medicines capable of modulating gene expression through Watson–Crick base pairing with their genetic targets. However, their clinical performance is strongly shaped by delivery barriers, including nuclease degradation, poor passive membrane permeability, rapid clearance and inefficient endosomal escape. To address these challenges, modern oligonucleotide drug design combines chemical stabilization with delivery-enabling conjugation strategies, such as phosphorothioate backbones, 2′-sugar modifications and ligand-directed targeting (2021; Roberts, Langer and Wood, 2020). Among these approaches, N-acetylgalactosamine (GalNAc) conjugation has become a leading strategy for hepatocyte delivery, as multivalent GalNAc ligands bind the asialoglycoprotein receptor (ASGPR) and promote receptor-mediated uptake into liver cells (Nair *et al*., 2014). Beyond GalNAc, RNA therapeutics are increasingly being engineered with lipids, peptides, antibodies, aptamers, polymers and nanoparticle systems to improve biodistribution, cellular uptake and tissue selectivity, expanding the potential of ASOs and other RNA-targeting agents beyond the liver while highlighting delivery as a central determinant of therapeutic index (Benizri *et al*., 2020; Dowdy, 2017; Roberts, Langer and Wood, 2020).

Feature	Small Molecules	GalNAc–siRNA Conjugates	Therapeutic Antibodies
1. Size	<1000 Da	~16,000 Da	>150,000 Da
2. Route of delivery	Oral, IV, IM, SC, inhalation	Subcutaneous (SC)	Mostly IV and SC
3. Plasma protein binding	Variable (high for lipophilic, low for hydrophilic)	High (~80–90%)	Low
4. Tissue distribution	Often large volume of distribution	Extensive liver distribution	Mostly confined to circulation & extracellular fluid
5. Metabolism	CYP450 enzymes (drug-metabolizing enzymes)	Exonucleases & endonucleases (no CYP involvement)	Proteolytic catabolism (receptor & non-specific)
6. Renal clearance	Often important pathway	Minor (<25% for intact siRNA)	None for intact antibodies
7. Plasma half-life	Short (hours)	Short (<10 hours)	Long (weeks)
8. PK linearity	Usually linear (can be nonlinear)	Linear	Often nonlinear at low doses (TMDD), linear at high doses
9. PK/PD relationship	PK → PD (PD depends on PK)	PK independent of PD; PD driven by RISC-loaded siRNA	PK often depends on PD (TMDD)

GalNAc

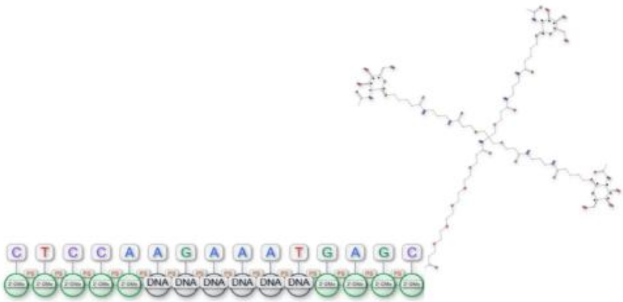
For many years, N-acetylgalactosamine, or GalNAc, was understood primarily as an amino sugar related to galactose. GalNAc helped form blood group A structures, appeared in mucin-type O-glycosylation, and marked important cancer-associated glycans such as the Tn antigen (Brockhausen and Stanley, 2015; Rodríguez et al., 2023). At this stage, GalNAc belonged mainly to glycobiology. It helped scientists understand how cells labelled themselves, communicated with their environment, and changed during disease.

In the 1960s at the National Institutes of Health, Gilbert Ashwell and Anatol Morell studied how the liver cleared circulating glycoproteins from the blood. They worked on plasma glycoprotein metabolism and noticed when glycoproteins lost terminal sialic acid and exposed galactose or GalNAc residues, the liver removed them rapidly. Their work led scientists to identify the hepatic asialoglycoprotein receptor (Ashwell and Morell, 1974; Stockert, 1995). This discovery gave GalNAc a new identity. Hepatocytes, the main functional cells of the liver, displayed the asialoglycoprotein receptor on their surface. That receptor recognized terminal galactose and GalNAc residues and internalized bound molecules. In practical terms, Ashwell and Morell had shown the field that the liver had a molecular doorway, and GalNAc could help unlock it.

This discovery became especially important when RNA medicines entered the scene. Small interfering RNAs, or siRNAs, could silence disease-causing genes, but they faced a delivery problem. They were large, charged, and vulnerable to degradation. Scientists could design powerful RNA molecules in the laboratory, but they still needed to get those molecules into the right cells in the body.

In the late 1990s, Erik A. L. Biessen and colleagues at the Division of Biopharmaceutics, Leiden Center for Drug Research, Leiden University, in the Netherlands, conjugated antisense oligodeoxynucleotides to a multivalent sugar ligand designed to bind the asialoglycoprotein receptor and redirect oligonucleotide uptake into parenchymal liver cells. This was an important proof of concept: an oligonucleotide that would otherwise be cleared or distributed inefficiently could be chemically attached to a hepatocyte-targeting ligand and pulled into the liver through receptor-mediated endocytosis (Biessen et al., 1999). These early glycoconjugates improved hepatocyte accumulation, but they did not yet produce the full pharmacological benefit that later GalNAc platforms would deliver. They showed both the promise and the problem: GalNAc-like receptor targeting could bring RNA drugs to the right tissue, but chemistry, linker design, oligonucleotide stability and intracellular trafficking still had to catch up.

Muthiah Manoharan and colleagues at Alnylam Pharmaceuticals then helped translate GalNAc conjugation into a delivery technology for RNAi therapeutics. Building these earlier studies, the Alnylam team developed multivalent GalNAc ligands that could be attached directly to chemically modified siRNAs, allowing the RNA duplex to reach hepatocytes after subcutaneous administration without the need for lipid nanoparticles. Their GalNAc-siRNA work showed that these conjugates localized to hepatocytes and produced robust RNAi-mediated gene silencing in vivo (Nair et al., 2014). Shortly after, T. P. Prakash and colleagues at Isis Pharmaceuticals took this principle to gapmer ASOs, showing that triantennary GalNAc conjugation improved ASO potency and hepatocyte delivery (Prakash et al., 2014). Together, these studies turned GalNAc from a glycobiology motif into one of the central delivery technologies for liver-targeted RNA medicines.



There are currently seven FDA-approved siRNA therapeutics on the market, reflecting the rapid clinical maturation of RNA interference as a therapeutic modality. Among these, patisiran remains the only approved siRNA drug formulated with a lipid nanoparticle (LNP) delivery system. In contrast, the other approved siRNA therapeutics, givosiran, lumasiran, inclisiran, vutrisiran, nedosiran, and fitusiran, use GalNAc conjugation to enable targeted delivery to hepatocytes. There are also approximately thirteen FDA-approved ASO therapeutics, including splice-modulating ASOs such as nusinersen and the Duchenne muscular dystrophy exon-skipping drugs eteplirsen, golodirsen, viltolarsen, and casimersen, as well as RNase H–mediated ASOs such as inotersen, tofersen, eplontersen, olezarsen, and donidalorsen. Unlike siRNA therapeutics, where GalNAc conjugation now dominates liver-directed delivery, most approved ASOs are not formulated in lipid nanoparticles; most are administered chemically “naked”, using backbone and sugar modifications to improve stability, potency, and tissue exposure. However, the field is increasingly moving toward targeted conjugation strategies: eplontersen was the first FDA-approved GalNAc-conjugated ASO, and newer liver-directed ASOs such as olezarsen also use GalNAc-based targeting.

Dicerna's GalXC platform

Dicerna's GalXC platform (acquired by Novo Nordisk) is a liver-targeted RNA interference (RNAi) technology designed to silence disease-associated genes at the mRNA level. The platform uses double-stranded RNA molecules conjugated to GalNAc, enabling selective uptake by hepatocytes through GalNAc-mediated receptor binding and internalization into liver cells (Dicerna Pharmaceuticals, 2021). Belcesiran, also known as DCR-A1AT, applied this platform to alpha-1 antitrypsin deficiency-associated liver disease (AATLD) by targeting hepatic production of alpha-1 antitrypsin (AAT), including the misfolded Z-AAT protein that accumulates in hepatocytes and contributes to liver injury, fibrosis, cirrhosis and hepatocellular carcinoma (Dicerna Pharmaceuticals, 2021).

The program was mechanistically supported by early clinical data. In a Phase I, double-blind, placebo-controlled, single-ascending-dose study in healthy volunteers, Belcesiran produced dose-dependent reductions in circulating serum AAT after subcutaneous administration. Interim data showed mean maximum serum AAT reductions of approximately 50%, 69% and 80% at the 1.0, 3.0 and 6.0 mg/kg dose levels, respectively, while individual participants in the 6.0 mg/kg cohort achieved maximum reductions ranging from 62% to 91% (Dicerna Pharmaceuticals, 2021). These results provided evidence of target engagement and supported progression into the Phase II ESTRELLA study. The early safety profile also appeared acceptable: no serious adverse events were reported in the interim Phase I analysis, most treatment-emergent adverse events were mild, and no clinically significant changes in laboratory parameters or lung function were observed during the treatment periods assessed (Dicerna Pharmaceuticals, 2021).

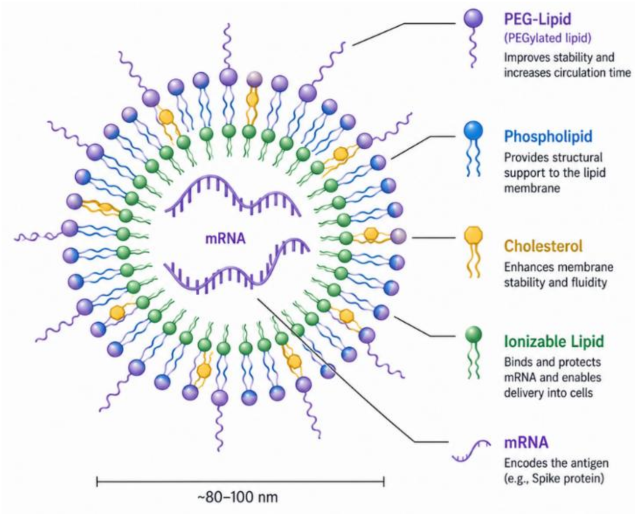
However, the subsequent Phase II ESTRELLA study did not generate the level of clinical evidence required to support continued development. ESTRELLA was designed as a randomized, double-blind, placebo-controlled, multiple-dose study evaluating the safety, tolerability, pharmacokinetics and pharmacodynamics of Belcesiran in adults with AATD-associated liver disease (ClinicalTrials.gov, 2026). The study was planned to include three cohorts, with the third cohort intended to expand the dataset and further characterize treatment effects, including liver-related outcomes. The trial was terminated early after enrolling only 16 participants, and the third cohort was never opened (ClinicalTrials.gov, 2026; Health Research Authority, 2026). The ClinicalTrials.gov record states that the study was terminated because the “development project was discontinued” (ClinicalTrials.gov, 2026).

Belcesiran does not appear to have failed because of an absence of pharmacodynamic activity. Repeated dosing continued to reduce blood AAT levels, suggesting that the RNAi mechanism remained active in the patient population (Health Research Authority, 2026). The more substantive issue was that AAT knockdown alone was insufficient to establish a compelling clinical path forward. For AATLD, the key development question is not simply whether circulating AAT can be reduced, but whether reducing hepatic Z-AAT production translates into meaningful liver benefit, such as reduced toxic protein accumulation, improvement or stabilization of fibrosis, or prevention of progression to cirrhosis and liver-related complications. Because ESTRELLA was stopped early, enrolled only a small number of participants and generated limited follow-up, it was unable to provide robust evidence on these disease-modifying outcomes.

Safety uncertainty may also have contributed to the cancellation of the program. Although the Health Research Authority summary described Belcesiran as generally safe and well tolerated overall, it also reported four serious adverse events considered possibly related to Belcesiran: pneumonia, atrial fibrillation, deterioration in lung function tests and worsening liver function, the latter explained by underlying cirrhosis due to AATLD. All affected participants recovered clinically (Health Research Authority, 2026). In a very small Phase II dataset, such events do not necessarily establish a causal safety signal, but they increase uncertainty, particularly for a therapy intended for patients who may already have clinically significant liver disease and who may also be vulnerable to pulmonary complications due to AAT deficiency.

Overall, the discontinuation of Belcesiran showed evidence of target engagement and had encouraging early tolerability data, but the curtailed Phase II study did not provide enough evidence that AAT lowering would translate into clinically meaningful liver benefit. (ClinicalTrials.gov, 2026; Health Research Authority, 2026).

	DRUG	COMPANY	TARGET	INDICATION
Rare Diseases	**GIVOSIRAN**	Alnylam	ALAS1	Acute hepatic porphyria
	LUMASIRAN	Alnylam	Glycolate oxidase	Primary hyperoxaluria type 1
	VUTRISIRAN	Alnylam	Transthyretin (TTR)	ATTR amyloidosis
	NEDOSIRAN	Novo Nordisk	LDHA	Primary hyperoxaluria type 1
	FAZIRSIRAN	Arrowhead	AAT (SERPINA1)	AAT deficiency liver disease
	FITUSIRAN	Alnylam / Sanofi	Antithrombin	Hemophilia
	SLN-124	Silence Therapeutics	TMPRSS6	Thalassemia
Cardiometabolic	**INCLISIRAN**	Alnylam / Novartis	PCSK9	Hypercholesterolemia
	OLPASIRAN	Amgen	Lp(a)	Cardiovascular disease
	ZILEBESIRAN	Alnylam	Angiotensinogen	Hypertension
	ZERLASIRAN	—	Lp(a)	Cardiovascular disease
	SOLBINSIRAN	Eli Lilly	ANGPTL3	Dyslipidemia / CV disease
Liver Diseases	**CEMDISIRAN**	Alnylam	Complement C5	Complement-mediated diseases
	ALN-HSD (ALN-HSD17B13)	Alnylam	HSD17B13	NASH
	BELCESIRAN	Alnylam / Dicerna	AAT	AAT liver disease
Infectious Diseases	**AB-729**	Arbutus	HBV proteins	Hepatitis B
	JNJ-3989	Arrowhead / J&J	HBV proteins	Hepatitis B
	ALN-HBV02	Alnylam	HBV proteins	Chronic HBV



Lipid nanoparticles (LNPs) are a clinically validated non-viral platform for delivering nucleic acid therapeutics, including mRNA and siRNA. Their clinical development has been driven by the success of COVID-19 mRNA vaccines and liver-directed RNA therapies, where LNPs have shown strong in vivo activity (Belaidi et al., 2025). Most LNPs contain four core components: an ionizable lipid, a helper phospholipid, cholesterol and a PEG-lipid. The ionizable lipid is especially important because it binds RNA during formulation, remains largely neutral in circulation and becomes protonated in the acidic endosome to support endosomal escape after cellular uptake (Belaidi et al., 2025). However, endosomal escape remains one of the main limits on LNP efficacy (Jozić et al., 2026; Belaidi et al., 2025).

Route of administration is a major determinant of LNP efficacy and safety. Intravenous administration is useful for systemic delivery, but it strongly directs LNPs toward the liver because circulating particles adsorb apolipoproteins, especially ApoE, and are taken up efficiently by hepatocytes and other liver-associated cells (Belaidi et al., 2025). This makes IV delivery well suited to liver diseases but less effective for extrahepatic targets such as the brain. For CNS therapies, the blood–brain barrier (BBB) adds another obstacle by limiting the entry of complex therapeutics, including nucleic acids, antibodies, immunotherapeutics and gene therapies (Iqbal, Hwang and Costain, 2026). As a result, CNS-directed LNPs often require additional engineering, such as optimized particle size, charge, lipid chemistry, surface ligands or receptor-mediated transcytosis strategies targeting receptors such as transferrin receptor, insulin receptor, IGF1R or LRP1 (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026).

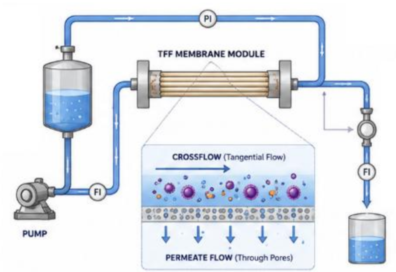
Different routes therefore create different safety concerns. Intramuscular administration, used mainly for vaccines, supports local antigen expression and immune activation rather than broad organ targeting. IV administration produces systemic exposure, making infusion-related reactions, complement activation, hepatic uptake and off-target biodistribution more relevant (Belaidi et al., 2025). Intrathecal, intracerebroventricular and intracerebral routes can improve CNS exposure by bypassing or reducing reliance on vascular transport barriers, but they introduce local inflammation, neurotoxicity, procedural burden and patient acceptability concerns (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026). Focused ultrasound with microbubbles offers a less invasive way to transiently open the BBB after IV dosing, but its safety may vary in diseased, inflamed or aged brain tissue (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026).

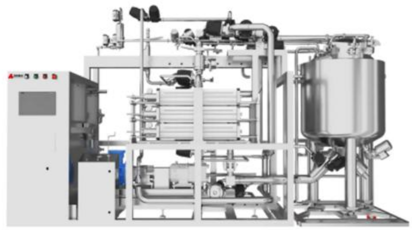
The toxicology profile of LNPs also depends on dose schedule. Acute toxicity is relatively well characterized and often includes transient flu-like symptoms, cytokine elevations and occasional infusion-related reactions. These effects depend on dose, infusion rate, lipid composition and patient context (Belaidi et al., 2025). Complement activation-related pseudo-allergy has been reported in high-infusion-rate animal models, and PEG-lipids remain under scrutiny because they may contribute to rare hypersensitivity-like responses (Belaidi et al., 2025).

Long-term toxicity is less well defined, especially for products that require repeated dosing. Chronic administration can promote anti-PEG antibody formation, accelerated blood clearance, altered biodistribution and reduced efficacy over time (Belaidi et al., 2025). Slowly metabolized lipids may also accumulate in the liver or other tissues, increasing the risk of chronic on-target and off-target toxicity. Biodegradable lipids are intended to reduce this risk by breaking down more readily after delivery, improving clearance and tolerability during repeat dosing (Belaidi et al., 2025). Overall, long-term safety must be evaluated in relation to route, target tissue, dosing frequency and disease context rather than generalized across all LNP products.

Manufacturing remains another major barrier to wider commercialization. LNP production requires reproducible large-scale manufacturing while maintaining critical quality attributes such as particle size, polydispersity, RNA encapsulation efficiency, sterility, stability and batch-to-batch consistency. This is difficult because LNPs self-assemble within milliseconds, and small changes in flow rate, mixing geometry, solvent dilution, shear stress, salt concentration, temperature or downstream purification can alter particle properties. As a result, commercial manufacturing must reproduce laboratory-scale performance across much larger volumes while preserving potency, biodistribution, stability and safety.

To manage these challenges, commercial platforms increasingly use technologies such as parallel impinging jet mixers, continuous-flow reactors and automated GMP systems. These require precise fluid-dynamic control, sterile closed-system designs, high-accuracy pumps and sensors, in-line monitoring and validated process controls. The cost burden is substantial, including GMP facilities, cleanrooms, continuous manufacturing systems, automated fill-finish lines, analytical instruments, sterile single-use consumables, expensive ionizable lipids, RNA payload losses and failed batches. For these reasons, manufacturing scalability remains one of the most important practical barriers to broad RNA-LNP commercialization.





For example, Tangential flow filtration (TFF) is commonly used to remove ethanol, exchange buffers, concentrate nanoparticles and prepare sterile formulations, but LNPs are fragile nanoscale structures. TFF can therefore cause shear-related particle damage, RNA leakage, membrane fouling, aggregation, flux decline and inconsistent product recovery. These failures can reduce encapsulation efficiency or compromise entire batches.

Overall, LNP manufacturing is not only a formulation challenge but also a fluid-dynamics and process-control problem. Successful production depends on millisecond-scale mixing, controlled solvent exchange, diffusion kinetics, shear management and precise regulation of self-assembly. Even minor deviations can affect product quality. For this reason, the field is increasingly investing in advanced mixing platforms, real-time monitoring, AI-driven process optimization and lyophilization to improve scalability, stability and reduce dependence on cold-chain logistics.

The complexity and cost of LNP therapeutics extend beyond manufacturing into storage, distribution and real-world deployment. Thermal instability and cold-chain dependence remain major limitations for RNA-LNP products, as the COVID-19 vaccine rollout demonstrated.